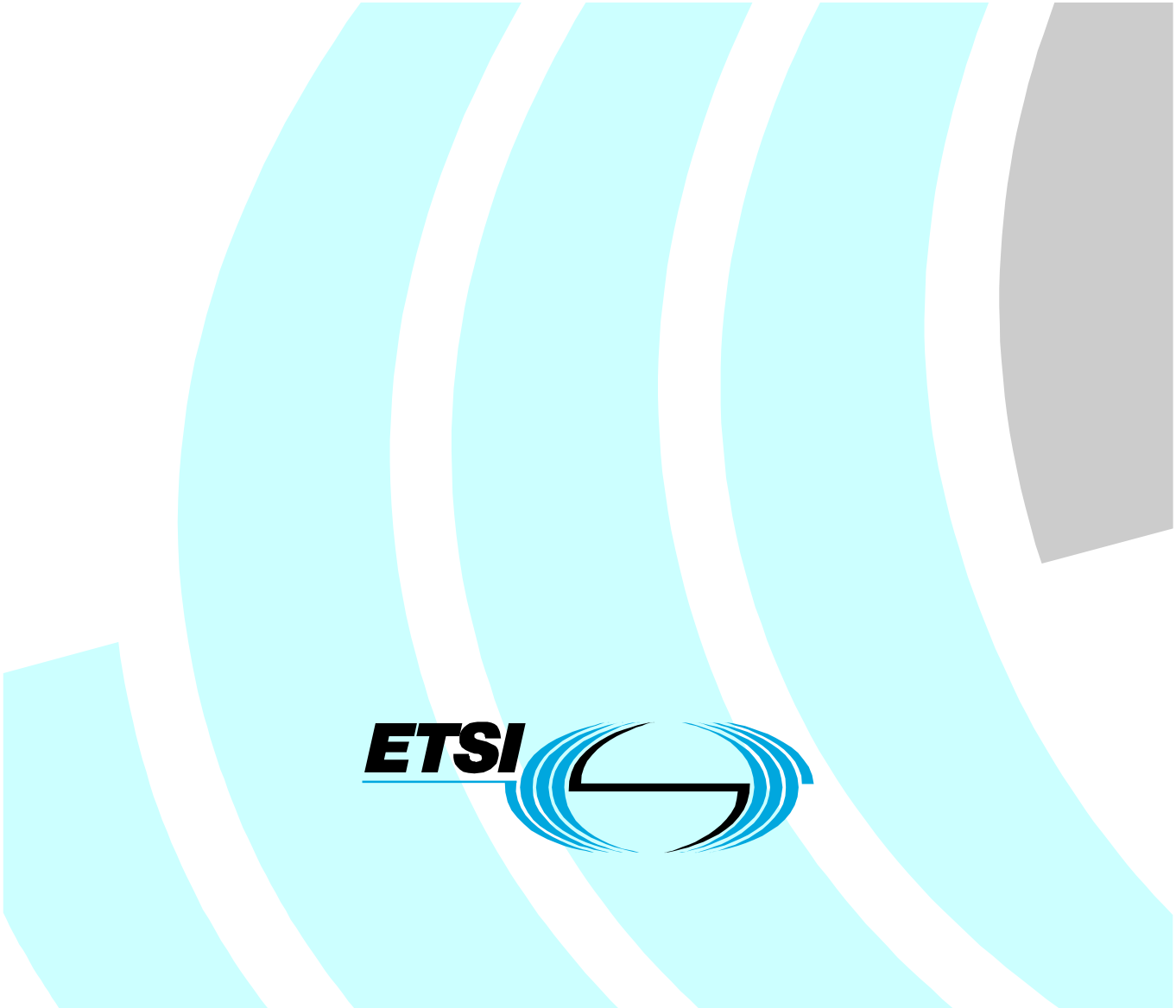
# ETSI ES 201 235-4 V1.3.1 (2006-03)

*ETSI Standard*

**Access and Terminals (AT);**
**Specification of Dual-Tone Multi-Frequency (DTMF)**
**Transmitters and Receivers;**
**Part 4: Transmitters and Receivers for use in**
**Terminal Equipment for end-to-end signalling**



Reference
RES/AT-010128

Keywords
DTMF, PSTN, signalling

***ETSI***

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

***Important notice***

Individual copies of the present document can be downloaded from:
http://www.etsi.org

The present document may be made available in more than one electronic version or in print. In any case of existing or perceived difference in contents between such versions, the reference version is the Portable Document Format (PDF). In case of dispute, the reference shall be the printing on ETSI printers of the PDF version kept on a specific network drive within ETSI Secretariat.

Users of the present document should be aware that the document may be subject to revision or change of status. Information on the current status of this and other ETSI documents is available at
http://portal.etsi.org/tb/status/status.asp

If you find errors in the present document, please send your comment to one of the following services:
http://portal.etsi.org/chaircor/ETSI_support.asp

***Copyright Notification***

No part may be reproduced except as authorized by written permission.
The copyright and the foregoing restriction extend to reproduction in all media.

© European Telecommunications Standards Institute 2006.
All rights reserved.

**DECT**™, **PLUGTESTS**™ and **UMTS**™ are Trade Marks of ETSI registered for the benefit of its Members.
**TIPHON**™ and the **TIPHON logo** are Trade Marks currently being registered by ETSI for the benefit of its Members.
**3GPP**™ is a Trade Mark of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners.

# Contents

Intellectual Property Rights ..... 4
Foreword ..... 4
1 Scope ..... 5
2 References ..... 5
3 Definitions, symbols and abbreviations ..... 5
3.1 Definitions ..... 5
3.2 Symbols ..... 6
3.3 Abbreviations ..... 6
4 The receiver ..... 6
4.1 Area of application ..... 6
4.2 Operate and non-operate requirements ..... 7
4.2.1 Signal condition requirements ..... 7
4.2.1.1 Valid signal for an analogue receiver ..... 7
4.2.1.2 Invalid signal for an analogue receiver ..... 7
4.2.1.3 Valid signal for a digital receiver ..... 8
4.2.1.4 Invalid signal for a digital receiver ..... 8
4.2.2 Digit Recognition Condition (DRC) requirements ..... 9
4.3 Speech immunity performance ..... 10
5 The transmitter ..... 10
6 Electrical characteristics ..... 10
**Annex A (normative): Speech immunity test signal ..... 11**
**Annex B (informative): Example test signal ..... 13**
**Annex C (informative): Bibliography ..... 14**
History ..... 15

# Intellectual Property Rights

IPRs essential or potentially essential to the present document may have been declared to ETSI. The information pertaining to these essential IPRs, if any, is publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI Web server (http://webapp.etsi.org/IPR/home.asp).

Pursuant to the ETSI IPR Policy, no investigation, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

# Foreword

This ETSI Standard (ES) has been produced by ETSI Technical Committee Access and Terminals (AT).

The present document is part 4 of a multi-part deliverable covering the Specification of Dual-tone Multi-Frequency (DTMF) Transmitters and Receivers, as identified below:

Part 1: "General";

Part 2: "Transmitters";

Part 3: "Receivers";

**Part 4: "Transmitters and Receivers for use in Terminal Equipment for end-to-end signalling".**

# 1 Scope

The present document specifies the Dual-tone Multi-Frequency (DTMF) signalling system.

The various parts of this multi-part deliverable provide a complete set of requirements for applications intending to use DTMF signalling.

The present document is intended to provide the level of detail that will enable manufacturers of telecommunications equipment incorporating DTMF signalling, to design the equipment such that it facilitates highly reliable signalling. This should not be taken to imply that any DTMF signalling system that fails to meet all the criteria described in the present document will not provide reliable signalling.

The present document is not intended to be used for the definition of regulated interfaces.

The present document covers the requirements for the DTMF transmitter and receiver for the case of DTMF signalling during an established call from TE to TE.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication and/or edition number or version number) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.

Referenced documents which are not found to be publicly available in the expected location might be found at http://docbox.etsi.org/Reference.

[1] ETSI ES 201 235 (all parts): "Access and Terminals (AT); Specification of Dual-tone Multi-Frequency (DTMF) Transmitters and Receivers".

[2] ETSI I-ETS 300 245-1: "Integrated Services Digital Network (ISDN); Technical characteristics of telephony terminals; Part 1: General".

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the following terms and definitions apply:

**Digit Recognition Condition (DRC):** output state of the receiver in response to a valid input signal when the relevant signal condition and time requirements are satisfied

**high group:** signalling frequencies, which have nominal values of 1 209 Hz, 1 336 Hz, 1 477 Hz and 1 633 Hz

**low group:** signalling frequencies, which have nominal values of 697 Hz, 770 Hz, 852 Hz and 941 Hz

**Valid Signal Condition (VSC):** state of the line at the input of the receiver when:

1) one of the frequency combinations specified in ES 201 235-1 is present; and
2) the frequencies are within the tolerances specified in clauses 4.2.1.1 and 4.2.1.3 item a); and

3) the signalling levels are within the range of acceptable levels specified in clauses 4.2.1.1 and 4.2.1.3 items b) and c); and

4) any unwanted frequencies are within the limits as specified in clauses 4.2.1.1 and 4.2.1.3 item i to iii) of ES 201 235-4

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

| | |
|---|---|
| * | The Star on the standard 3x4 keypad array, also known as the asterisk |
| # | The Square on the standard 3x4 keypad array, also known as the hash, sharp, or number sign ("pound" in the USA) |
| dBm0 | Power level expressed in decibels relative to 1 mW in a 0 dBr level reference point |
| dBV | Absolute voltage level expressed in decibels relative to 1 volt |

## 3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| CLI | Calling Line Identification |
| DRC | Digit Recognition Condition |
| DTMF | Dual-tone Multi-Frequency |
| HPF | High Pass Filter |
| LPF | Low Pass Filter |
| R | Rectifiers |
| SPF | Single Pole Filter |
| STF | Single Tone Filter |
| TC | Threshold Comparator |
| TE | Terminal Equipment |
| VSC | Valid Signal Condition |

# 4 The receiver

## 4.1 Area of application

The present document deals with the receiver or receiver function for DTMF signalling incorporated into TE. This implies that the receiver needs to be capable of dealing with both the frequency related attenuation as a result of two local loops and the exchanges and interconnecting trunk circuits as well as any changes in frequency that might be introduced by the equipment used by the network provider(s).

The receiver shall respond to signals received over a network from a distant transmitter conforming to clause 5.

The receiver shall be equipped to recognize the sixteen signals designated 0 to 9, *, #, A, B, C and D. The system may be utilized with only, ten discrete signals, in which case the signals designated 0 to 9, should be used; or with twelve discrete signals, in which case the signals 0 to 9, *, # should be used. In the interests of system security, the receiver may be conditioned to recognize only specific digits from the complete set of sixteen digits.

NOTE: The principal uses of the receiver as dealt with in the present document are expected to be:

(a) in TE receiving signals from a remote TE e.g. remote control of an answering/recording machine;

(b) in TE receiving signals from equipment such as an CLI based server. This may be connected so as not to introduce the extra attenuation of the second local connection.

# 4.2 Operate and non-operate requirements

## 4.2.1 Signal condition requirements

The following clauses define the conditions under which the DTMF function shall decide to indicate or not indicate to the host equipment the digit that is represented by a received signal. There will be some conditions that fall into neither of these two states and the designer of the DTMF function can make their own decision whether such signals are best considered valid or not valid.

### 4.2.1.1 Valid signal for an analogue receiver

The received DTMF signal shall be detected as valid when only two of the signalling frequencies are present at the receiver input one from the high group and one from the low group which meet the conditions described in items a) to c):

a) each of these signalling frequencies is within ± (1,5 % +2 Hz) of the nominal value; and

b) the level of each of these two signalling frequencies is within the range -43 dBV to -3 dBV; and

c) the difference in level of these two signalling frequencies is not more than 10 dB.

The received DTMF signal shall also be detected as valid even if accompanied by other frequency components not exceeding:

i) in the frequency range 300 Hz to 480 Hz a level of -43 dBV;

ii) in the frequency range 480 Hz to 3 400 Hz a total voltage level of 20 dB below the level of the low group signalling frequency;

iii) at frequencies over 3 400 Hz, an individual signal level of -36 dBV at 3 400 Hz rising at 6 dB per octave to a level of +10 dBV.

NOTE 1: In item b), a dynamic range of 40 dB is specified to take into account tolerances in the send levels of the DTMF transmitter and also variations in attenuation found in connections via various networks.

NOTE 2: It is pointed out that some equipment may require protection to cover interference from specified sources, e.g. meter pulses. This may be in addition to that specified in item iii).

### 4.2.1.2 Invalid signal for an analogue receiver

The received DTMF signal shall be considered as invalid if any of the conditions given in items a) to c) are satisfied at the receiver input:

a) the signal levels are correct for valid signals but more than two of the signalling frequencies are present; or

b) the level of the signalling frequency in the low group is less than -48 dBV; or

c) the level of the signalling frequency in the high group is less than -48 dBV.

As an aid to the understanding of clauses 4.2.1.1 and 4.2.1.2, table 1 is given.

**Table 1: Signal condition requirements**

| Signal condition | Frequency deviation (Hz) | Signal level x (dBV) | Signal level difference y (dB) | Unwanted frequencies z (dBV) |
|---|---|---|---|---|
| **Is valid** | ≤ ± (1,5 % + 2) | -43 ≤ x ≤ -3 | y < 10 | z ≤ Ψ |
| **May be valid** | > ± (1,5 % + 2) | -48 ≤ x < -43<br>or<br>x > -3 | y ≥ 10 | z > Ψ |
| **Invalid** | | x < -48 | | |
| NOTE 1: Ψ see clause 4.2.1.1 items i) to iii).<br>NOTE 2: It is realized that certain electrical signals may exist on the line at the input of the receiver which do not fulfil either the requirements of the "***Is valid***" row or those of the "***Invalid***" row. In these circumstances, the requirements for the existence or non-existence of signal condition are not specified directly, but indirectly, through the speech immunity requirements of clause 4.3. | | | | |

### 4.2.1.3 Valid signal for a digital receiver

The received DTMF signal shall be detected as valid when only two of the signalling frequencies are present at the receiver input, one from the high group and one from the low group which meet the conditions described in items a) to c):

a) each of these signalling frequencies is within ± (1,5 % +2 Hz) of the nominal value; and

b) the level of each of these two signalling frequencies is within the range -35 dBm0 to -4 dBm0; and

c) the difference in level of these two signalling frequencies is not more than 6 dB.

The received DTMF signal shall also be detected as valid even if accompanied by other frequency components not exceeding:

i) in the frequency range 15 Hz to 480 Hz; an individual signal level of 0 dBm0 at 15 Hz to 50 Hz with the limit then falling at 12 dB per octave until 300 Hz. From 300 Hz to 480 Hz a level of -31 dBm0;

ii) in the frequency range 480 Hz to 3 400 Hz a total voltage level of 20 dB below the level of the low group signalling frequency;

iii) at frequencies over 3 400 Hz, an individual signal level of -36 dBm0 at 3 400 Hz rising at 6 dB per octave to a level of 0 dBm0.

NOTE 1: In item b), a dynamic range of 31 dB is specified to take into account tolerances in the send levels of the DTMF transmitter and also variations in attenuation found in the local lines of the various networks.

NOTE 2: It is pointed out that some networks may require protection to cover interference from specified sources, e.g. meter pulses. This may be in addition to that specified in item iii).

### 4.2.1.4 Invalid signal for a digital receiver

The received DTMF signal shall be considered as invalid if any of the conditions given in items a) to c) are satisfied at the receiver input:

a) the signal levels are correct for valid signals but more than two of the signalling frequencies are present; or

b) the level of the signalling frequency in the low group is less than -40 dBm0; or

c) the level of the signalling frequency in the high group is less than -40 dBm0.

As an aid to the understanding of clauses 4.2.1.3 and 4.2.1.4, table 2 is given.

**Table 2: Signal condition requirements**

| Signal condition | Frequency deviation (Hz) | Signal level x (dBm0) | Signal level difference y (dB) | Unwanted frequencies z (dBm0) |
|---|---|---|---|---|
| **Valid** | ≤ ± (1,5 % + 2) | -35 ≤ x ≤ -4 | y < 8 | z ≤ Ψ |
| **May be valid** | > ± (1,5 % + 2) | -40 ≤ x < -35 or x > -4 | y ≥ 8 | z > Ψ |
| **Not valid** | | x < -40 | | |
| NOTE 1: Ψ see clause 4.2.1.3 items i) to iii).<br>NOTE 2: It is realized that certain electrical signals may exist on the line at the input of the receiver which do not fulfil either the requirements of the "***Is valid***" row or those of the "***Is not valid***" row. In these circumstances, the requirements for the existence or non-existence of signal condition are not specified directly, but indirectly, through the speech immunity requirements of clause 4.3. | | | | |

## 4.2.2 Digit Recognition Condition (DRC) requirements

If a valid signal present at the DTMF receiver input fulfils the conditions set out in items a) and b) then the digit shall be indicated as recognized:

a) the signal condition shall be preceded by the continuous absence of a valid signal condition for more than 40 ms or (optionally) by a recognized digit that is different to the current digit; and

b) the signal condition shall then exist continuously for more than 40 ms.

If a valid signal at the receiver input has been present for less than 20 ms, the digit shall not be indicated as recognized.

Once the DRC exists, it shall be unaffected by disturbances having a duration of less than 20 ms.

Thus a detected digit that subsequently has interruptions that are less than 20 ms, shall be interpreted as a single digit.

As an aid to the understanding of clause 4.2.2, table 3 is given.

**Table 3: Digit recognition condition requirements**

| Character recognition | When ***not DRC*** * and then |
|---|---|
| **Shall exist** | VSC > 40 ms |
| **May exist** | 20 ms ≤ VSC ≤ 40 ms |
| **Shall not exist** | VSC < 20 ms |

| Character recognition | When ***DRC*** and then |
|---|---|
| **Shall cease** | Not VSC > 40 ms |
| **May cease** | 20 ms ≤ Not VSC ≤ 40 ms |
| **Shall not cease** | Not VSC < 20 ms |

* or optionally a different digit recognition condition

Legend:
- VSC — The continuous existence of a valid signal condition.
- Not VSC — The continuous non-existence of valid signal condition.
- DRC — The existence of digit recognition condition.
- Not DRC — The non-existence of digit recognition condition.

## 4.3 Speech immunity performance

The receiver should not perform more than 46 false operations per 100 speech hours at a long-term active speech level of -13 dBV at analogue receivers and -13 dBm0 for digital receivers.

A speech hour is defined as a total duration of one hour when speech (in both directions) may, be present, that is when the circuit is seized and the call established (conversation condition).

NOTE 1: In addition to the simulation of signals by speech entering via the microphone, mis-operation may also be caused by disturbances preventing detection of valid signals and/or splitting them into double digits. The design of a receiver with very high rejection efficiency against signal imitation might lead to an intolerable increase in the number of blockings. The general design aim should be to limit both types of mis-operation (imitation and blocking) to a total level not exceeding one per 10 000 digits dialled.

NOTE 2: The duration of 100 hours has been given as the recommended test duration but sufficient accuracy can be achieved by using 10 speech hours, which can be further reduced by applying the time compression procedure described in annex A.

For test of the speech immunity performance or for comparison between different receivers the test specified in annex A can be used.

# 5 The transmitter

The analogue transmitters for use in terminal equipment for end-to-end signalling shall comply with the requirements as defined in part 2 of this multi-part deliverable.

The digital transmitters for use in terminal equipment for end-to-end signalling shall comply with the requirements as defined in part 2 of this multi-part deliverable with the following amendment:

- The sending levels for each frequency, measured selectively, shall be -15 dBm0 with the tolerance of ±2 dB.

For this end-to-end signalling option in ISDN telephony terminals, see I-ETS 300 245-1 [2].

# 6 Electrical characteristics

NOTE: The electrical characteristics of the interface will be determined by the apparatus into which the function of the DTMF receiver or transmitter has been incorporated.

# Annex A (normative): Speech immunity test signal

The speech immunity performance of the receiver is tested by means of a standardized common test signal.

The common standardized test signal is a collection of speech "bursts" which simulate characters on a reference receiver when applied at the NTP. The reference unprotected receiver is defined in figure A.1 and in the notes to figure A.1.

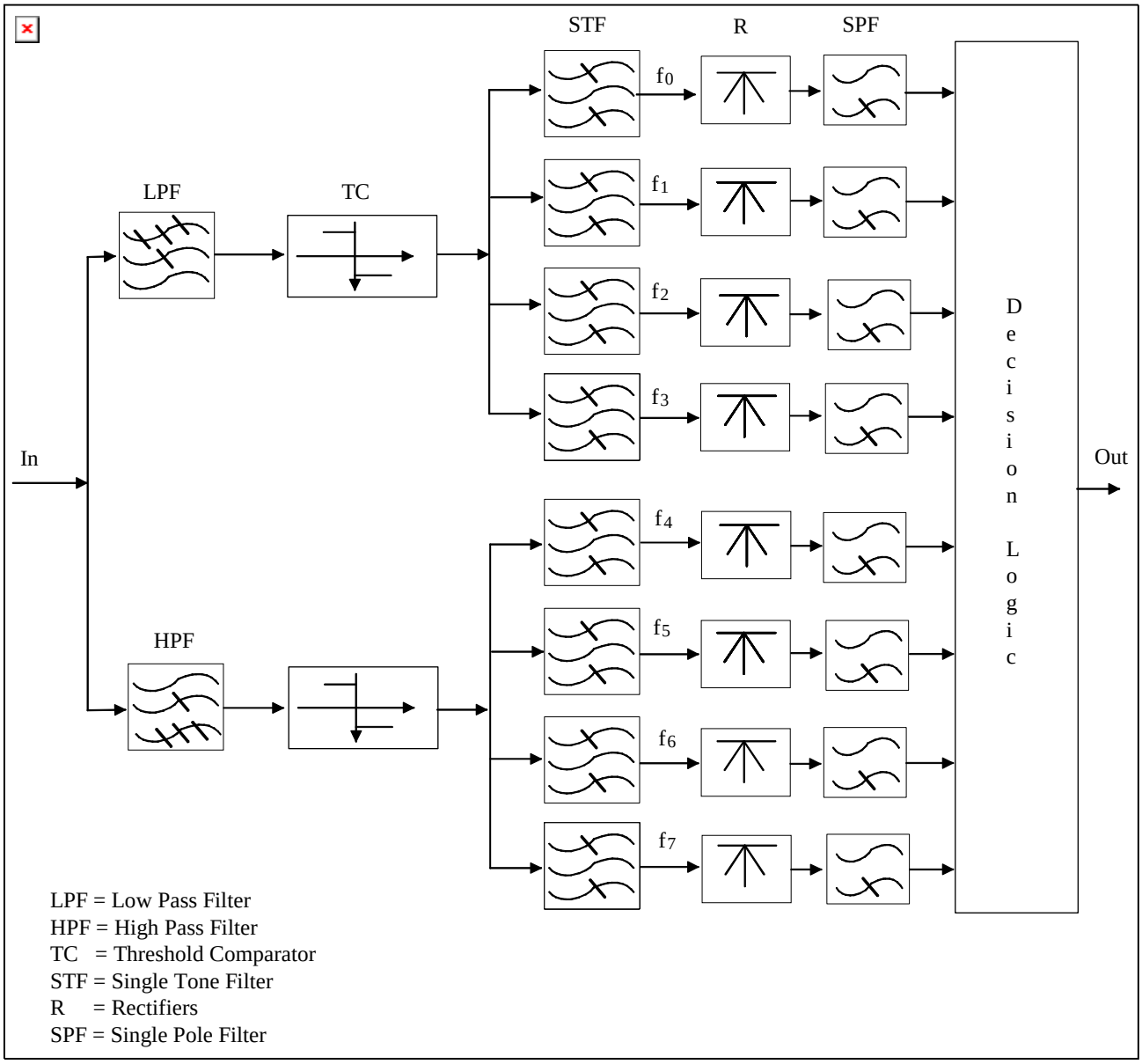
Each time the receiver recognizes a false digit, the corresponding section of speech signal is sampled, the sampling window length being in the range from 100 ms to 200 ms, equally distributed before and after the false digit recognition instant.

The duration of the pauses included between the sampled bursts shall range between 150 ms and 300 ms.

The original speech source used for generating the test sequence shall include speech samples from at least four of the following languages: English, French, German, Italian and Spanish, and shall contain both female and male voices.

Based on the available experience, the processing of 10 hours of speech material results in a test signal of about 20 minutes. When this test signal is used, no more than 5 false operations shall occur when it is applied to the DTMF receiver.

The test sequence shall include a 1 kHz calibration tone with a level 12 dB higher than the active speech level of the speech burst sequence.




NOTE 1: The scheme of the reference unprotected exchange receiver is given in figure A.1.
The characteristics of the components are:

- low-pass filter f ≤ 960 Hz;
- high-pass filter f ≥ 1 190 Hz;
- two threshold comparators;
- eight channel filters (two pole filters: -1,5 dB band pass limits at the nominal frequency ± (1,5 % + 2 Hz));
- eight rectifiers;
- eight single pole (RC) filters, time constant C = 4 ms to 5 ms.

NOTE 2: The thresholds and decision logic are regulated in order not to recognize character signals with a level below -30 dBm and to recognize valid character signals with a level in the range of -4 dBm to -30 dBm.

NOTE 3: No other protections are to be provided in the receiver.

**Figure A.1: Reference unprotected exchange receiver (Block diagram)**

# Annex B (informative): Example test signal

One example for a test signal suitable for DTMF receivers is available from Telecom Italia Lab.

For further information please contact:

Telecom Italia Lab
LAP - Accredited Testing Laboratory
Via Guglielmo Reiss Romoli, 274
10148 TORINO
ITALY
http://lap.telecomitalialab.com
e-mail: lap@tilab.com

# Annex C (informative): Bibliography


- ITU-T Recommendation Q.23: "Technical features of push-button telephone sets".
- ITU-T Recommendation Q.24: "Multifrequency push-button signal reception".
- ITU-T Recommendation P.57: "Artificial Ears".

# History

| Document history | | |
|---|---|---|
| V1.1.1 | September 2000 | Publication |
| V1.2.1 | May 2002 | Publication |
| V1.3.1 | January 2006 | Membership Approval Procedure MV 20060310: 2006-01-10 to 2006-03-10 |
| V1.3.1 | March 2006 | Publication |
| | | |